

# Tsai-Shien Chen

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## Education

### University of California, Merced

Merced, CA

#### PH.D. CANDIDATE IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Sep. 2022 - Now

- Advisor: Prof. Mukesh Singhal
- GPA: 4.00 / 4.00

### National Taiwan University

Taipei, Taiwan

#### M.S. IN ELECTRONICS ENGINEERING

Sep. 2019 - March 2022

- Advisor: Prof. Shao-Yi Chien
- GPA: 4.30 / 4.30; Rank: 1st / 128

#### B.S. IN ELECTRICAL ENGINEERING

Sep. 2015 - June 2019

- GPA: 4.23 / 4.30; Rank: 5th / 190

## Research Interests

- Video Generation: Dataset, Foundation model, and Application
- Controllable and Personalized Visual Content Creation

## Honors & Awards

- 2026 **CVPR Outstanding Reviewers (top 5% of reviewers)**
- 2026 **CVPR Doctoral Consortium**
- 2025 **Google PhD Fellowship**
- 2024 **Graduate Student Opportunity Program Fellowship (one year of full tuition and stipend), UC Merced**
- 2023 **Best Master Thesis of the Year Award**, Graduate Institute of Electronics Engineering, NTU
- 2022 **Honorary Member (top 2% in college)**, Phi Tau Phi Scholastic Honor Society
- 2019 **Valedictorian**, Department of Electrical Engineering, NTU
- 2015-2019 **4-time Presidential Award (top 5% in department)**, NTU

## Work Experiences

### Adobe Research

San Francisco, CA

#### RESEARCH INTERN AT CREATIVE INTELLIGENCE LAB

March 2026 - Now

- Mentors: Richard Zhang, Eli Shechtman

### Snap Inc.

Santa Monica, CA

#### RESEARCH INTERN AT CREATIVE VISION TEAM

May 2023 - March 2026

- Mentors: Aliaksandr Siarohin, Sergey Tulyakov, Jun-Yan Zhu, Kfir Aberman
- Large-scale video dataset and video captioning, generation, and personalization models.

## Publications

### Conference Proceedings

- Canvas-to-Image: Compositional Image Generation with Multimodal Controls**  
Yusuf Dalva, Gordon Guocheng Qian, Maya Goldenberg, **Tsai-Shien Chen**, Kfir Aberman, Sergey Tulyakov, Pinar Yanardag, Kuan-Chieh Jackson Wang  
SIGGRAPH 2026
- Omni-Attribute: Open-vocabulary Attribute Encoder for Visual Concept Personalization**  
**Tsai-Shien Chen**, Aliaksandr Siarohin, Gordon Guocheng Qian, Kuan-Chieh Jackson Wang, Egor Nemchinov, Moayed Haji-Ali, Riza Alp Guler, Willi Menapace, Ivan Skorokhodov, Anil Kag, Jun-Yan Zhu, Sergey Tulyakov  
CVPR 2026
- AlcheMinT: Fine-grained Temporal Control for Multi-Reference Consistent Video Generation**  
Sharath Girish, Viacheslav Ivanov, **Tsai-Shien Chen**, Hao Chen, Aliaksandr Siarohin, Sergey Tulyakov  
CVPR 2026

4. **Multi-subject Open-set Personalization in Video Generation**  
**Tsai-Shien Chen**, Aliaksandr Siarohin, Willi Menapace, Yuwei Fang, Kwot Sin Lee, Ivan Skorokhodov, Kfir Aberman, Jun-Yan Zhu, Ming-Hsuan Yang, Sergey Tulyakov  
*CVPR 2025*
5. **VIMI: Grounding Video Generation through Multi-modal Instruction**  
Yuwei Fang, Willi Menapace, Aliaksandr Siarohin, **Tsai-Shien Chen**, Kuan-Chieh Jackson Wang, Ivan Skorokhodov, Graham Neubig, Sergey Tulyakov  
*EMNLP 2024*
6. **Panda-70M: Captioning 70M Videos with Multiple Cross-Modality Teachers**  
**Tsai-Shien Chen**, Aliaksandr Siarohin, Willi Menapace, Ekaterina Deyneka, Hsiang-wei Chao, Byung Eun Jeon, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Ming-Hsuan Yang, Sergey Tulyakov  
*CVPR 2024*
7. **Snap Video: Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**  
Willi Menapace, Aliaksandr Siarohin, Ivan Skorokhodov, Ekaterina Deyneka, **Tsai-Shien Chen**, Anil Kag, Yuwei Fang, Aleksei Stoliar, Elisa Ricci, Jian Ren, Sergey Tulyakov  
*CVPR 2024 (Highlight Presentation)*
8. **Incremental False Negative Detection for Contrastive Learning**  
**Tsai-Shien Chen**, Wei-Chih Hung, Hung-Yu Tseng, Shao-Yi Chien, Ming-Hsuan Yang  
*ICLR 2022*
9. **Hard Samples Rectification for Unsupervised Cross-domain Person Re-identification**  
Chih-Ting Liu, Man-Yu Lee, **Tsai-Shien Chen**, Shao-Yi Chien  
*ICIP 2021*
10. **Tracklet-Refined Multi-Camera Tracking Based on Balanced Cross-Domain Re-Identification for Vehicles**  
Kai-Siang Yang, Yu-Kai Chen, **Tsai-Shien Chen**, Chih-Ting Liu, Shao-Yi Chien  
*CVPR Workshops 2021*
11. **Orientation-aware Vehicle Re-identification with Semantics-guided Part Attention Network**  
**Tsai-Shien Chen**, Chih-Ting Liu, Chih-Wei Wu, Shao-Yi Chien  
*ECCV 2020 (Oral Presentation)*
12. **Viewpoint-Aware Channel-Wise Attentive Network for Vehicle Re-Identification**  
**Tsai-Shien Chen**, Man-Yu Lee, Chih-Ting Liu, Shao-Yi Chien  
*CVPR Workshops 2020*
13. **Supervised Joint Domain Learning for Vehicle Re-Identification**  
Chih-Ting Liu, Man-Yu Lee, Chih-Wei Wu, Bo-Ying Chen, **Tsai-Shien Chen**, Yao-Ting Hsu, Shao-Yi Chien  
*CVPR Workshops 2019*

## Pre-prints

1. **CineOrchestra: Unified Entity-Centric Conditioning for Cinematic Video Generation**  
Sharath Girish\*, **Tsai-Shien Chen\***, Zhikang Dong, Mukesh Singhal, Hao Chen, Sergey Tulyakov, Aliaksandr Siarohin  
*Coming Soon*
2. **LayerComposer: Multi-Human Personalized Generation via Layered Canvas**  
Gordon Guocheng Qian, Ruihang Zhang, **Tsai-Shien Chen**, Yusuf Dalva, Anujraaj Argo Goyal, Willi Menapace, Ivan Skorokhodov, Meng Dong, Arpit Sahni, Daniil Ostashev, Ju Hu, Sergey Tulyakov, Kuan-Chieh Jackson Wang  
*arXiv preprint 2025*

## Patents

1. **Captioning Videos with Multiple Cross-Modality Teachers**  
**Tsai-Shien Chen**, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Aliaksandr Siarohin, Sergey Tulyakov  
*US12489956B2*
2. **Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**  
**Tsai-Shien Chen**, Yuwei Fang, Anil Kag, Willi Menapace, Jian Ren, Aliaksandr Siarohin, Ivan Skorokhodov, Sergey Tulyakov  
*US20250245897A1*

## Professional Activities

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| Invited Talks       | <ul style="list-style-type: none"><li>- <b>ByteDance</b> (April 2026): “From Data to Director: Towards Controllable and Personalized Movie Generation”</li><li>- <b>AI Time</b> (Dec 2025): “From Personalization Models towards Movie Generator”</li><li>- <b>Anuttacon</b> (July 2025): “Learning Video Generation and Personalization from Large-Scale Datasets”</li></ul> |
| Program Committee   | <ul style="list-style-type: none"><li>- <b>AAAI</b>: 2026</li></ul>                                                                                                                                                                                                                                                                                                           |
| Conference Reviewer | <ul style="list-style-type: none"><li>- <b>CVPR</b>’22,’23,’24,’25,’26; <b>ICCV</b>’21,’23,’25; <b>ECCV</b>’22,’24,’26; <b>WACV</b>’26; <b>ACCV</b>’24</li><li>- <b>NeurIPS</b>’23,’24,’25,’26; <b>ICLR</b>’24,’25,’26; <b>ICML</b>’24,’25,’26</li><li>- <b>SIGGRAPH</b>’26; <b>SIGGRAPH Asia</b>’26; <b>ACMMM</b>’25,’26</li></ul>                                           |
| Journal Reviewer    | <ul style="list-style-type: none"><li>- IEEE Transactions on Pattern Analysis and Machine Intelligence (<b>TPAMI</b>)</li><li>- IEEE Transactions on Image Processing (<b>TIP</b>)</li><li>- International Journal of Computer Vision (<b>IJCV</b>)</li></ul>                                                                                                                 |
| Teaching Assistant  | <ul style="list-style-type: none"><li>- UC Merced: CSE-185 <b>Introduction to Computer Vision</b> (Spring 2023)</li><li>- UC Merced: CSE-024 <b>Advanced Programming</b> (Fall 2022)</li><li>- NTU: EEE-5053 <b>Computer Vision</b> (Spring 2021)</li></ul>                                                                                                                   |